

EBill – Digital Receipt Generator (DRG)

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled “ **EBILL GENERATOR**” is the bonafide work of **Safeekur Rahman (220701236)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This project presents a comprehensive digital solution for enhancing transaction management through the development of an eBill system — a MERN (MongoDB, Express.js, React, Node.js) based Digital Receipt Generator (DRG) aimed at improving efficiency, reducing environmental impact, and strengthening security in modern commerce. Unlike traditional paper-based receipt systems, the proposed platform digitizes the entire receipt lifecycle, from generation to long-term storage, offering users a seamless and accessible interface for tracking expenses and managing transactions. The DRG system is built to support features such as real-time receipt generation, secure cloud-based storage, and detailed reporting tools that allow businesses and customers to analyze their financial activities through monthly and yearly summaries.

A key innovation in this project lies in its integration of analytics features that provide insights into purchasing behavior, transaction patterns, and business performance. The platform compares favorably to paper receipts in terms of environmental sustainability, cost-efficiency, and data accessibility. With growing concerns about paper waste and manual accounting errors, eBill addresses these challenges by offering a scalable and eco-conscious alternative that ensures both privacy and reliability.

In addition to its technical foundation, the system emphasizes usability and adaptability across diverse business environments, from retail to service-based industries. Future work will explore the integration of AI-driven analytics for spending prediction, multi-user dashboards, and real-time synchronization across multiple platforms to further enhance decision-making and financial planning.

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LIST OF ABBREVIATIONS

S. No	ABBR	Expansion
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	AJAX	Asynchronous JavaScript and XML
4	ASGI	Asynchronous Server Gateway Interface
5	AWT	Abstract Window Toolkit
6	BC	Block Chain
7	CSS	Cascading Style Sheet
8	DFD	Data Flow Diagram
9	DSS	Digital Signature Scheme
10	GB	Gradient Boosting
11	JSON	JavaScript Object Notation
12	ML	Machine Learning
13	RF	Random Forest
14	SQL	Structure Query Language
15	SVM	Support Vector Machine

CHAPTER 1

INTRODUCTION

1.1 GENERAL

In the rapidly evolving landscape of digital transformation, businesses and consumers are increasingly seeking efficient, secure, and environmentally conscious alternatives to traditional paper-based processes. One such process, the generation and storage of transaction receipts, continues to rely heavily on physical paper in many industries, despite its drawbacks such as high maintenance costs, susceptibility to damage or loss, and environmental burden due to excessive paper consumption. With growing global emphasis on sustainability, operational efficiency, and data-driven decision-making, there is a pressing need for systems that not only digitize receipts but also enhance the way transactional data is recorded, stored, and analyzed.

The limitations of paper receipts extend beyond inconvenience and waste. They lack portability, pose security risks due to loss or tampering, and contribute to inefficient accounting practices. Moreover, businesses often struggle to aggregate and analyze transaction data for financial reporting, business intelligence, and customer behavior insights. In the age of digital payments and cloud infrastructure, such challenges signal an urgent call for a paradigm shift in how transactional records are handled.

Digital Receipt Generators (DRGs) offer a viable solution to this problem by providing electronic alternatives that are secure, accessible, cost-effective, and environmentally friendly. However, many existing systems fall short in functionality, failing to provide detailed analytics, real-time accessibility, and long-term data management. In this

context, a more robust, intelligent, and user-friendly solution is required — one that not only replaces paper but adds strategic value to business operations.

With the emergence of the MERN (MongoDB, Express.js, React, Node.js) stack, it is now possible to create scalable, full-stack web applications capable of supporting complex data workflows in real time. The proposed eBill project leverages this stack to create an integrated system that not only generates digital receipts but also provides advanced features such as monthly and yearly financial summaries, centralized data storage, and detailed analytical dashboards for users and businesses alike

1.2 OBJECTIVE

In light of the growing need for sustainable, intelligent, and efficient transaction management systems, the primary objective of this project is to design and implement **eBill** — a MERN-based **Digital Receipt Generator** (DRG) that reimagines how receipts are created, managed, and utilized in both consumer and business contexts. The goal is not merely to digitize receipts, but to deliver a comprehensive web-based platform that enhances financial clarity, data security, and environmental responsibility.

At its core, eBill replaces traditional paper receipts with secure, cloud-stored digital versions that users can access from any device at any time. But beyond basic digitization, the project introduces key innovations such as the automatic generation of **monthly and yearly account summaries**, enabling users and business owners to track spending trends, analyze transaction patterns, and maintain organized financial records effortlessly. By transforming receipts into structured, queryable data, eBill also enables integration with analytical tools for deeper business insights.

To address varying user needs and maximize adoption, the system is built with a focus on **user-friendly interfaces**, **scalability**, and **cross-platform accessibility**. It ensures that small businesses, retailers, and consumers can all benefit from a seamless, reliable, and eco-conscious solution. Furthermore, the project examines the **cost-benefit advantages** of digital receipts over paper-based alternatives, demonstrating tangible improvements in operational efficiency and environmental impact.

Ultimately, the project aims to pave the way for smarter, data-centric financial ecosystems. Future enhancements may include AI-driven financial forecasting, real-time budget notifications, and integration with payment gateways and accounting software to create a truly intelligent and adaptable financial management tool.

1.3 EXISTING SYSTEM

Traditional billing and receipt systems across commercial and retail sectors have largely remained paper-based or only partially digitized. In such environments, receipts are printed and handed over physically to customers — a method that is not only resource-intensive but also prone to loss, wear, and limited usability beyond the point of sale. These receipts are often discarded or misplaced, leading to inefficiencies in personal expense tracking, business auditing, and long-term financial planning. Even in businesses that employ point-of-sale (POS) systems, the digitization typically stops at invoice generation, without a holistic system for receipt storage, retrieval, or analysis.

While some modern businesses have started offering digital alternatives — such as emailed receipts or SMS-based confirmations — these solutions are usually fragmented and lack centralized access or analytics capabilities. Customers may receive isolated emails for each transaction, but they are rarely offered tools to categorize, track, or

analyze their spending over time. Similarly, businesses receive limited insight from these basic systems, often having to rely on external accounting software for performance analysis.

A few existing platforms do provide receipt storage or expense tracking apps (e.g., Expensify, Zoho Expense, or Wave), but they often cater to corporate use or require integration with complex financial systems. These platforms usually charge subscription fees and are geared toward enterprises rather than small businesses or individual users. Moreover, they tend to emphasize finance and expense reporting rather than transactional interaction and eco-friendly digitization.

Critically, most of these systems lack a unified, user-friendly interface that caters to both customers and businesses, offering real-time access to receipts, detailed transaction analytics, and customized reporting features. Additionally, very few platforms prioritize sustainability by eliminating paper usage entirely or offer features such as **monthly/yearly summaries**, **visual data dashboards**, and **automated insights** without requiring technical expertise.

Security, too, is a concern in many current systems, with insufficient encryption or data handling standards leading to potential vulnerabilities in financial data. These shortcomings underline the pressing need for a more inclusive, intelligent, and scalable digital receipt solution that aligns with the technological expectations and environmental responsibilities of the modern world.

CHAPTER 2

LITERATURE SURVEY

1. “A Survey on Electronic Billing Systems: Trends and Technologies” – K. S. Reddy et al. (2018)

This paper reviews the development of electronic billing platforms, highlighting cloud integration, receipt digitization, and eco-friendly benefits. It discusses security challenges and the rise of web-based billing services, setting the context for platforms like DRG.

2. “Digital Receipts: Enhancing Customer Experience and Business Intelligence” – A. Mehta & S. Bhatt (2020)

This study explores the use of digital receipts for improving customer service and transactional transparency. It emphasizes how real-time e-receipts combined with data analytics can help businesses understand consumer patterns.

3. “The Role of Cloud Storage in E-Invoicing Systems” – L. Zhou et al. (2019)

Focusing on cloud-based billing systems, this work details how MongoDB and other NoSQL databases can securely store and retrieve receipt data, offering scalability and reliability over traditional storage.

4. “Environmental Impact of Paper Receipts vs. Digital Alternatives” – J. Carter & M. Lee (2017)

This paper compares the ecological footprint of physical vs. digital receipts, concluding that eBill systems drastically reduce paper waste, making them a sustainable solution aligned with global green initiatives.

5. “Secure Receipt Generation Using Node.js and Express Frameworks” – P. Singh & R. Thomas (2021)

The authors propose a secure backend architecture using Node.js and Express.js for generating and managing receipts. It discusses encryption methods and session handling to prevent fraud and unauthorized access.

6. “Data Visualization for Transaction Analysis in Web Applications” – K. Banerjee et al. (2020)

This paper presents methods for integrating reporting tools like Chart.js or D3.js with React interfaces to display monthly/yearly summaries. It supports DRG’s goal of actionable insights via visualization.

7. “MERN Stack for Modern Web Applications: A Case Study” – S. Rao & D. Kapoor (2022)

A practical case study demonstrating how MongoDB, Express, React, and Node can be used together to build full-stack systems. It highlights component-based UI design, RESTful APIs, and asynchronous data handling.

8. “Blockchain and Security in Digital Billing Systems” – Y. Tanaka & H. Lee (2019)

While not MERN-specific, this paper addresses how blockchain or tokenized verification can enhance trust in digital receipts. It explores integration possibilities with cloud-based billing services.

9. “Smart Retail: Integrating Digital Receipt Systems with POS Platforms” – T. Sharma & L. Hu (2018)

This research discusses how e-receipts can be directly linked to Point-of-Sale (POS) systems, offering seamless transaction documentation and instant customer notifications—features relevant to DRG.

10. “User Experience Design for Financial Dashboards” – M. Feldman & I. Krueger (2021)

This study evaluates UX design principles for financial tracking apps, stressing clarity, responsiveness, and real-time updates—key to making DRG adaptable across industries like retail and services.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

The proposed system, **eBill – Digital Receipt Generator (DRG)**, is a full-stack web-based application designed using the **MERN stack** (MongoDB, Express.js, React, Node.js) to offer a sustainable, secure, and intelligent alternative to traditional paper-based billing systems. This system addresses the limitations of conventional and semi-digital receipt models by introducing a centralized, automated platform that not only generates digital receipts in real time but also provides advanced analytics for business intelligence.

At the core of the system is a **modular architecture** that separates frontend presentation, backend logic, and database storage. The **frontend**, built in React, offers a responsive and intuitive interface for generating, viewing, and emailing receipts. The **backend**, developed with Express.js and Node.js, handles API requests, authentication, data processing, and routing. All transactional and customer data are securely stored in **MongoDB**, enabling flexible and scalable document-based data management.

A key innovation of this system is the **automated monthly and yearly report generation module**, which provides detailed summaries of sales, expenses, and revenue trends. This module supports export to common formats such as PDF or Excel and can automatically email reports to the business owner, reducing manual accounting effort and enabling data-driven decision-making. The analytics engine allows business owners to track performance trends, identify top-selling items, and monitor transaction frequency, making it an invaluable tool for both micro and small enterprises.

The system also emphasizes **data security and user privacy** by incorporating modern web security practices such as JWT-based authentication, role-based access control, and encrypted database entries. This ensures that both customers and businesses can trust the platform with sensitive transaction data. Additionally, the system is **cloud-ready** and can be deployed on services like Render, Vercel, or Heroku for real-time accessibility across devices.

To further enhance usability, features such as **receipt search and filter**, **customer email notifications**, and **date-wise transaction viewing** have been integrated. These user-centric capabilities ensure that businesses can retrieve, manage, and share receipts effortlessly without relying on manual filing or paper documentation.

By combining **cost-efficiency, sustainability, scalability, and intelligent features**, the eBill Digital Receipt Generator offers a modern, future-ready solution for billing and transaction management. Unlike existing receipt systems that focus solely on digital output, this system incorporates **business analytics, automated communication, and eco-friendly practices**, bridging the gap between traditional finance workflows and next-generation technology.

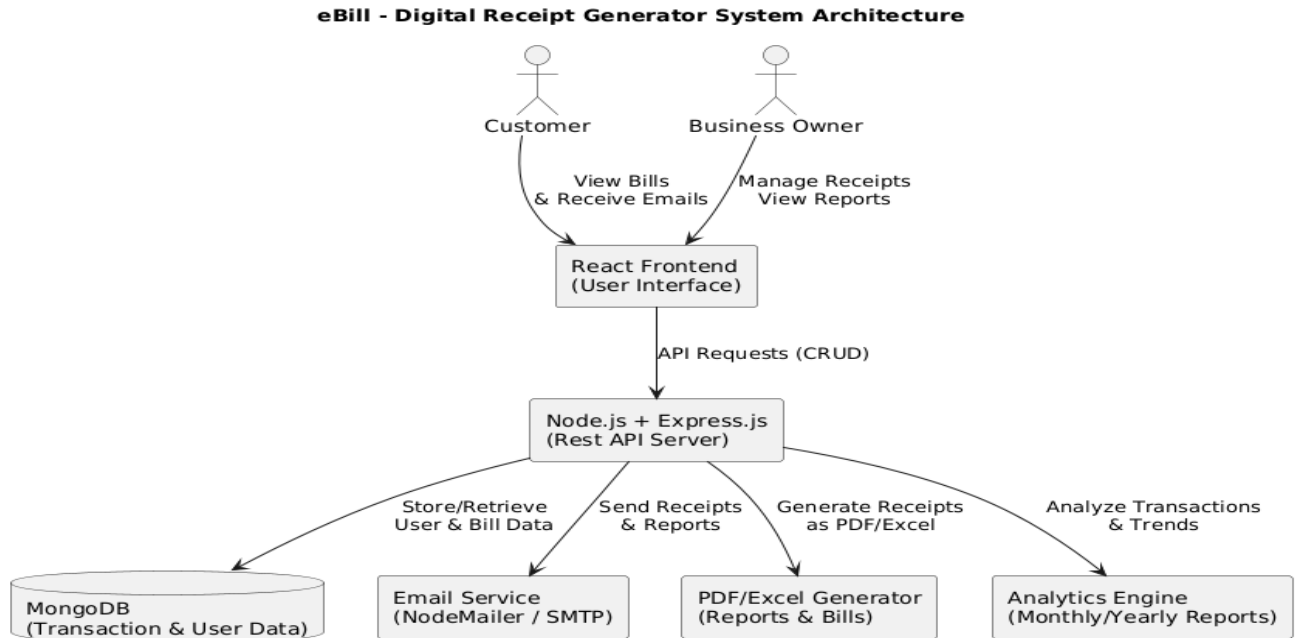


Fig 3.1: System Architecture

3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

The hardware specifications could be used as a basis for a contract for the implementation of the system. This therefore should be a full, full description of the whole system. It is mostly used as a basis for system design by the software engineers.

Table 3.1 Hardware Requirements

COMPONENTS	SPECIFICATION
------------	---------------

PROCESSOR	Intel Core i3
RAM	4 GB RAM
POWER SUPPLY	+5V power supply

3.3.2 SOFTWARE REQUIREMENTS

The software requirements paper contains the system specs. This is a list of things which the system should do, in contrast from the way in which it should do things. The software requirements are used to base the requirements. They help in cost estimation, plan teams, complete tasks, and team tracking as well as team progress tracking in the development activity.

Table 3.2 Software Requirements

COMPONENTS	SPECIFICATION
Operating System	Windows 7 or higher
Frontend	ReactJS,CSS
Backend	Express js
Database	MongoDB

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1 ACTIVITY DIAGRAM

The activity diagram for the Digital Receipt Generator begins with the **user login or registration** activity, ensuring secure access. Once authenticated, the user proceeds to the **transaction initiation** activity, where purchase details are entered. This triggers the **receipt generation** process, where the input data is processed using the backend (Node.js and Express.js) and stored in **MongoDB**. After successful data storage, a **digital receipt is generated and displayed** to the user through the React frontend. The user then has the option to **download, email, or archive** the receipt. Periodically, the system performs **monthly and yearly report generation**, allowing for **data analytics and business insights**. The process ends with the user either logging out or initiating a new transaction..

Activity Diagram - Digital Receipt Generator (eBill)

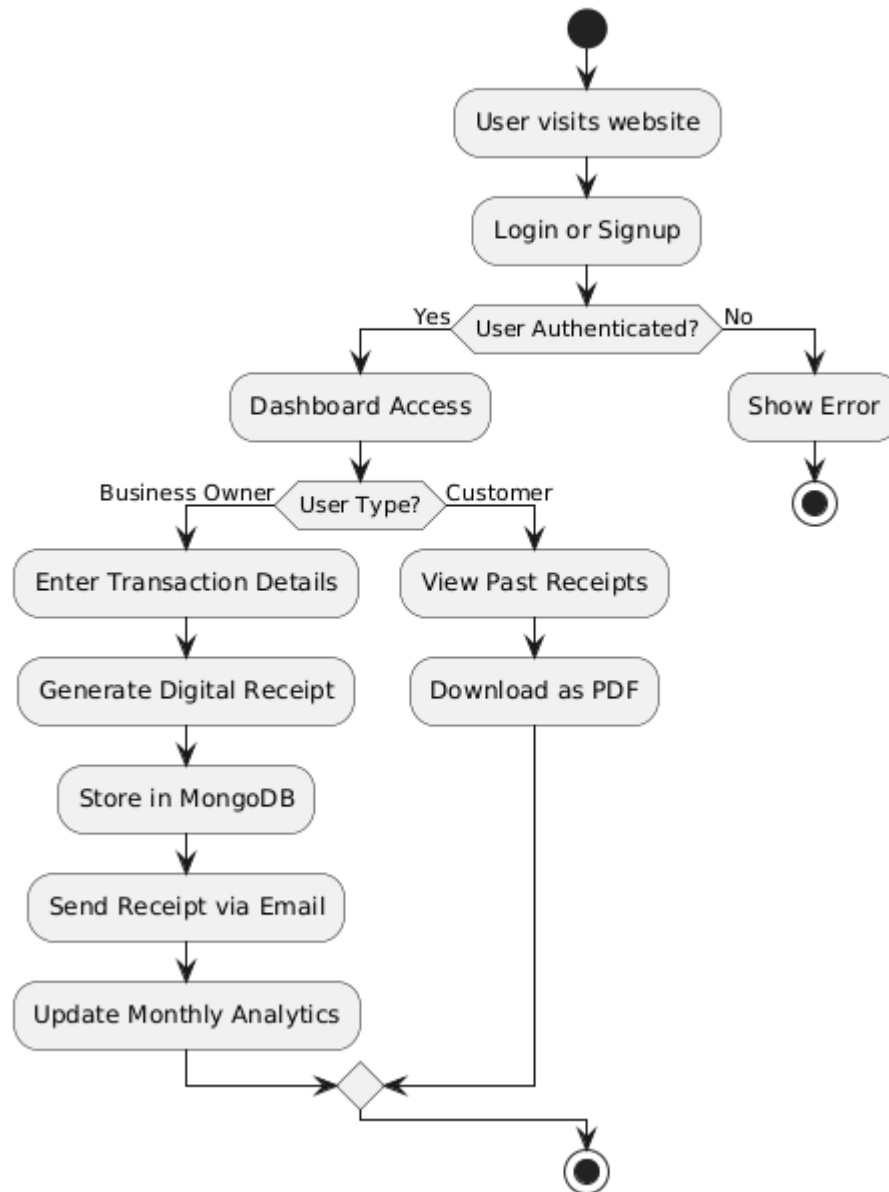


Fig 3.2: Activity Diagram

3.4.2 DATA FLOW DIAGRAM

The Data Flow Diagram begins with the **User** as the external entity who interacts with the system by entering login credentials and transaction details. These inputs are sent to the **Frontend Interface (React.js)**, which validates and forwards the data to the **Backend Server (Node.js and Express.js)**. The backend processes this data and interacts with the **Database (MongoDB)** to store transaction and receipt information securely. Once stored, the backend sends a confirmation and receipt data back to the frontend, which then **generates and displays the digital receipt** to the user. Additionally, scheduled data is fetched from the database for **monthly and yearly report generation**, and is processed to provide **account summaries and business analytics**. The results are presented back to the user through the frontend for viewing or exporting.

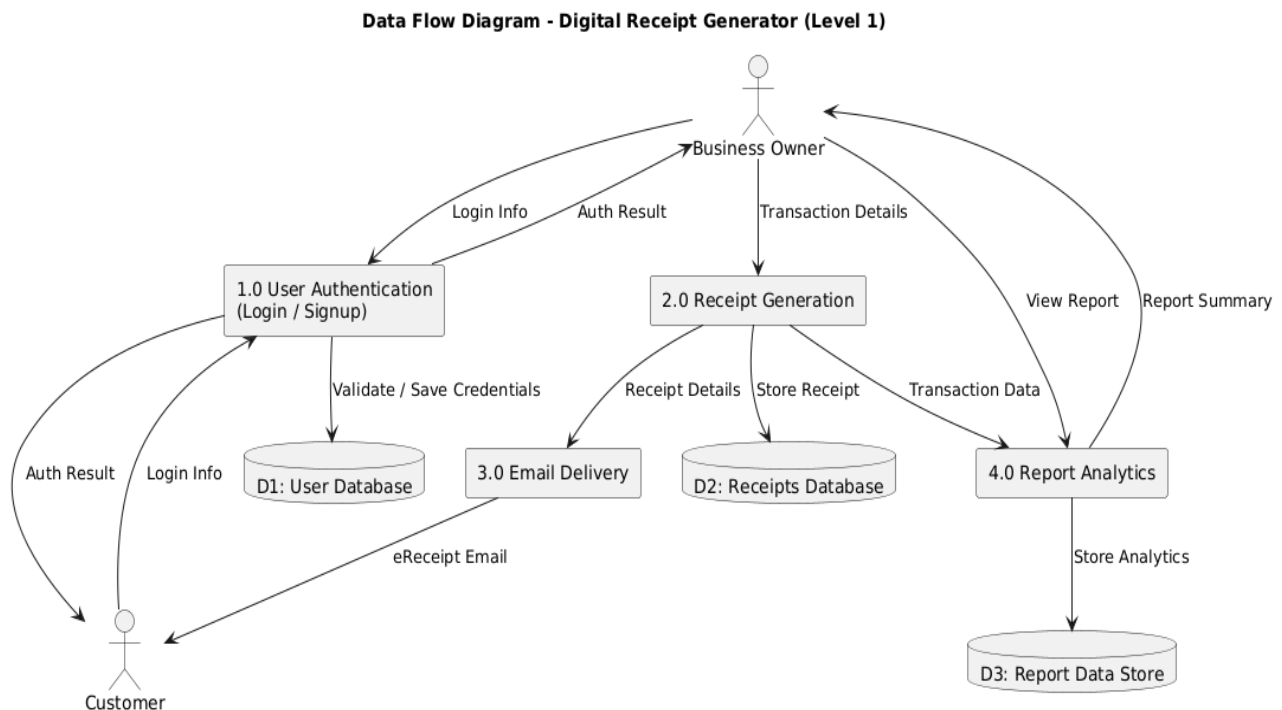


Fig 3.3:Data Flow Diagram

3.5 STATISTICAL ANALYSIS

1. Security:

- **Paper Receipts:** Vulnerable to being lost, forged, or tampered with. Physical storage can lead to unauthorized access.
- **Digital Receipts:** Securely stored with encryption and access control. Easily backed up and cannot be lost unless data is deleted or corrupted.

Hypothetical Comparison:

- Loss or forgery rate for paper receipts: 10%
- Loss or forgery rate for digital receipts: 1%

2. Accessibility:

- **Paper Receipts:** Require physical storage, can be misplaced or damaged over time.
- **Digital Receipts:** Can be accessed anytime, anywhere with internet connectivity, and can be searched easily in a digital archive.

Hypothetical Comparison:

- Paper receipts accessible within 5 minutes: 50%
- Digital receipts accessible anytime (via app/website): 95%

3. Cost:

- **Paper Receipts:** Include costs for paper, printing, ink, and storage.

- **Digital Receipts:** The initial cost of setting up the digital system but very low ongoing costs after that.

Hypothetical Comparison:

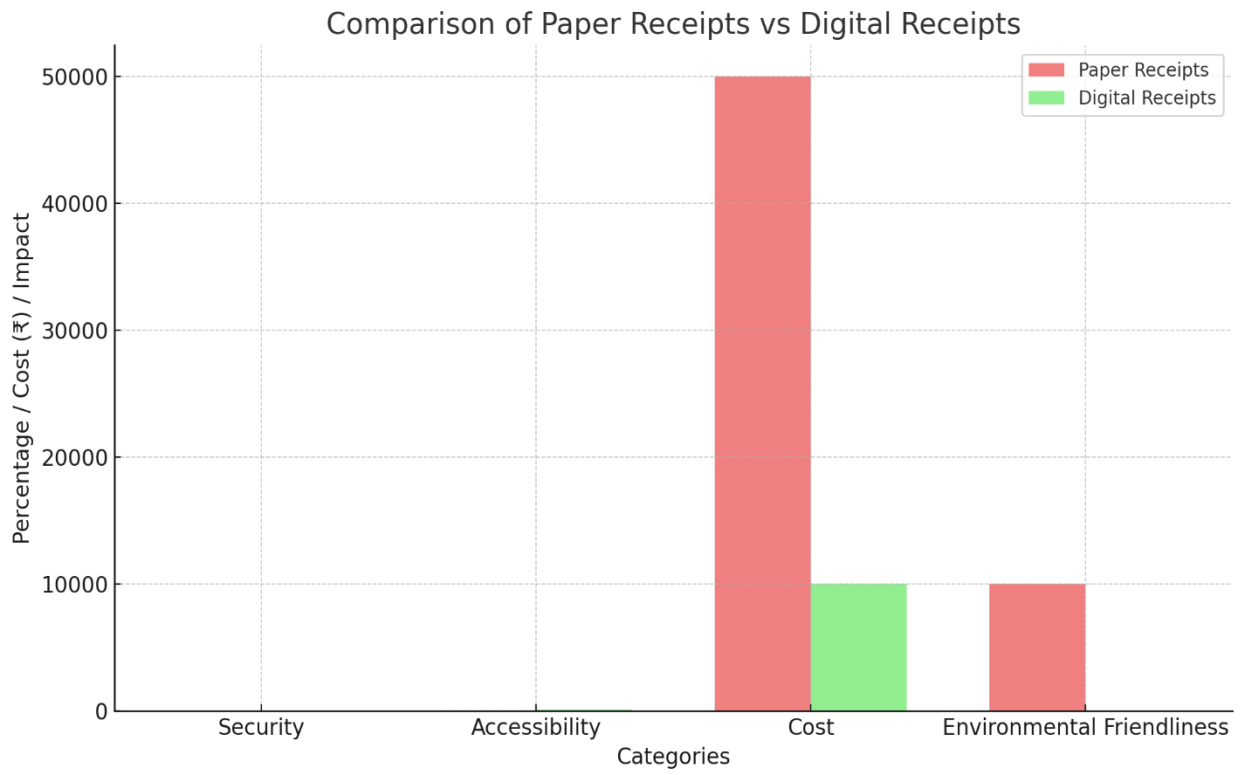
- Annual cost for paper receipts per business: ₹50,000
- Annual cost for digital receipts per business (system maintenance, storage): ₹10,000

4. Environmental Friendliness:

- **Paper Receipts:** Contribute to deforestation, energy use in production, and paper waste.
- **Digital Receipts:** Reduce paper waste, but require energy for servers and devices.

Hypothetical Comparison:

- Paper receipts generated per year (100 businesses): 10,000,000
- Digital receipts generated per year (same 100 businesses): 0 (eco-friendly)

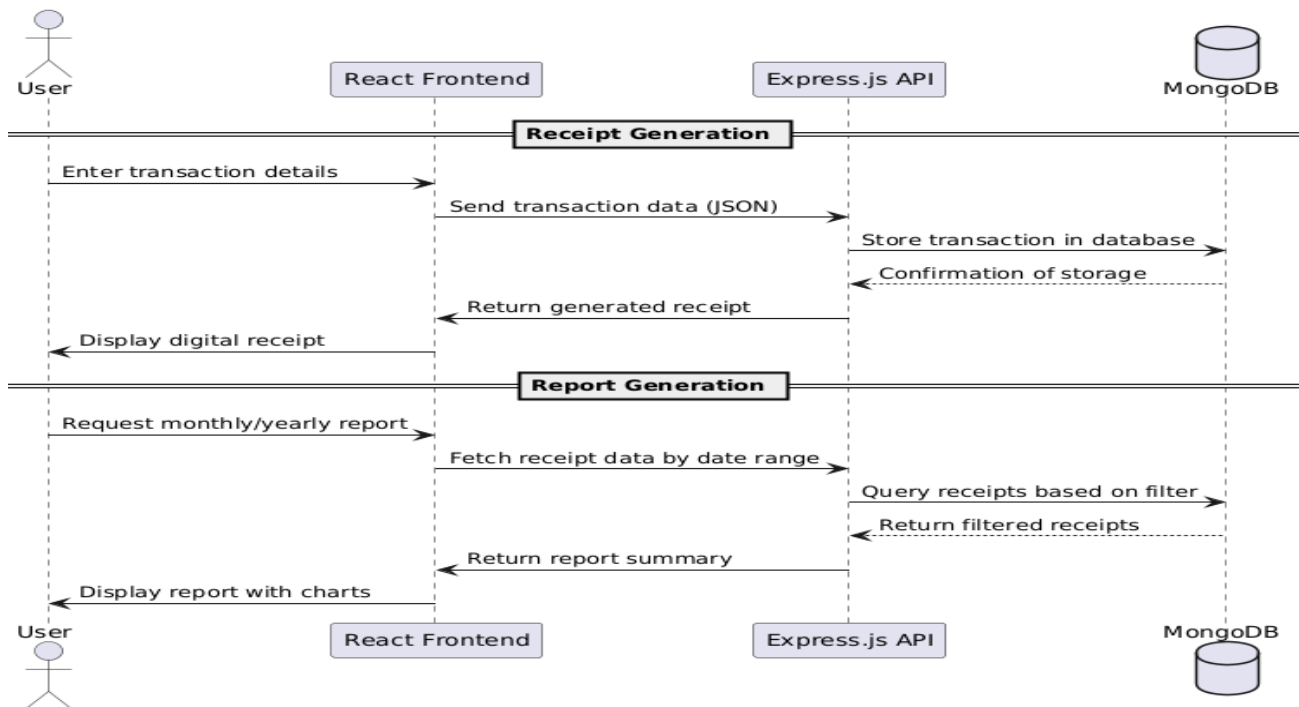


CHAPTER 4

MODULE DESCRIPTION

4.1.1 USER INTERFACE DESIGN

The sequence diagram outlines the interaction between the user and the DRG system. The process starts when a user accesses the web interface and inputs transaction details. This information is sent to the backend server via secure APIs. The server processes the input, stores it in the MongoDB database, and then generates a structured digital receipt. The generated receipt is then returned to the frontend and displayed to the user. The user has options to download, email, or view the receipt history. For reports, the user selects a date range, and the system fetches relevant transaction data, computes totals, and visualizes it using graphs and summaries.



Sequence diagram

4.1.2 BACK END INFRASTRUCTURE

The backend is built with Node.js and Express.js, responsible for handling requests, validating input, and interacting with MongoDB. The system ensures secure storage of receipt data with timestamps, amounts, and metadata. Monthly and yearly report generation logic is implemented server-side to summarize and group receipts based on dates. Additional modules allow for real-time updates, error logging, and optional user authentication. The backend also handles exporting receipts and reports in PDF/CSV formats for easy sharing or archiving.

CHAPTER 5

5.1 OUTPUT SCREENSHOTS

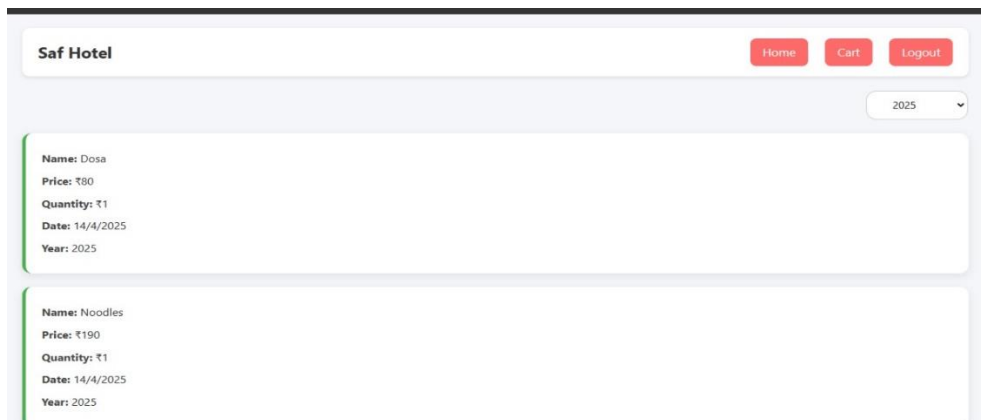
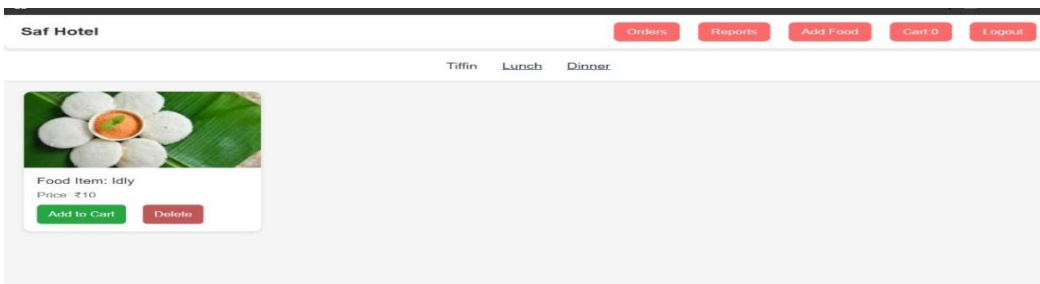
Login

Email

Password

Login

Donot have an account? [Sign up](#)



Add food

Food name

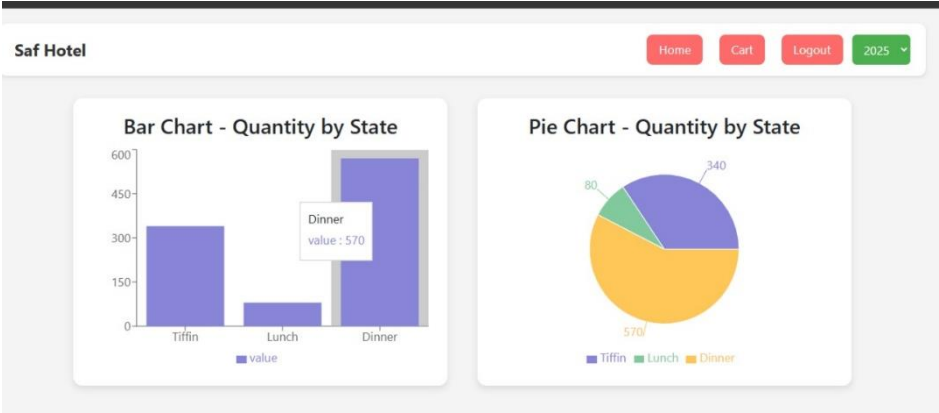
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--Type--

ADD

Back to home



Your Cart

Idly Qty: 1 ₹ 10 Delete

Meals Qty: 1 ₹ 80 Delete

Total: ₹ 90

Email

Send Bill

1 of 866

Your Hotel Bill - Ace Spad Hotel

Inbox x

220701236@rajalakshmi.edu.in

to me

7:15 PM (0 minutes ago)

☆

😊

↶

⋮

Thank you for dining with us. Please find your bill attached.

One attachment

Scanned by Gmail

ACE SPAD HOTEL

NEAR XYZ ROAD, CITY, PIN-123456

S.No	Item Name	Qty	Price	Total
1	Idly	1	110	110
2	Meals	1	80	80
			Grand Total	190

Thanks for dining with us!

AceSpadHotel_Bil...

↶ Reply

↷ Forward

😊

ACE SPAD HOTEL				
Near XYZ Road, City, PIN-123456				
Bill To: azsafrah@gmail.com			Date: 4/22/2025	
S.No	Item Name	Qty	Price	Total
1	Idly	1	110	110
2	Meals	1	80	80
			Grand Total	190
Thank you for dining with us!				

CHAPTER -6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The Digital Receipt Generator (e-Bill Generator) project effectively replaces traditional paper receipts with secure, eco-friendly, and cost-effective digital alternatives. By leveraging the MERN stack (MongoDB, Express.js, React, Node.js), the system provides a scalable and seamless solution for generating and managing digital receipts in real-time. The integration of data analytics allows users to easily access detailed reports, offering a comprehensive view of their transactions and financial activity.

This system ensures ease of use with a responsive frontend built using Tailwind CSS, allowing users to interact intuitively with the platform. The backend, powered by Node.js and Express, efficiently handles user requests, stores receipt data securely in MongoDB, and generates insightful reports for the users.

The environmentally friendly approach, paired with cost-effective implementation, enhances the value of the system, making it a viable alternative to traditional receipt methods. By providing users with the ability to manage and track their receipts, and generate monthly or yearly reports, the e-Bill Generator significantly improves financial tracking and accountability.

In conclusion, this project paves the way for a more sustainable and efficient method of managing receipts, promoting not only environmental benefits but also user convenience and financial transparency.

6.2 Future Enhancements

1. To further streamline the process, the system could integrate with popular payment gateways (like PayPal, Stripe, or Razorpay). This would enable the automatic generation of digital receipts directly from transactions, eliminating manual input and enhancing convenience for both businesses and customers.
2. To cater to an international audience, adding support for multiple currencies and automatic conversion based on the user's location could be a valuable feature. This would allow businesses operating globally to generate receipts in their local currencies.
3. As the number of receipts generated increases, it would be beneficial to include a powerful search and filter function. Users could search receipts by keywords, categories, dates, or amounts, making it easier to find specific transactions or summaries of their expenses.
4. Developing a mobile version of the e-Bill Generator would make the system more accessible on the go. A mobile app could allow users to generate, store, and manage receipts directly from their smartphones, providing more flexibility and convenience.
5. Expanding the analytics capabilities to provide more detailed insights into spending patterns, trends, and personalized recommendations could add significant value. Integrating machine learning for predictive analytics, such as forecasting future spending or identifying irregularities in transaction data, could further enhance the system's usefulness.

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